COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 2.2

Verification of Even and Odd Signal Decomposition

Aim:

Decompose a signal into even and odd parts and verify reconstruction.

Program:

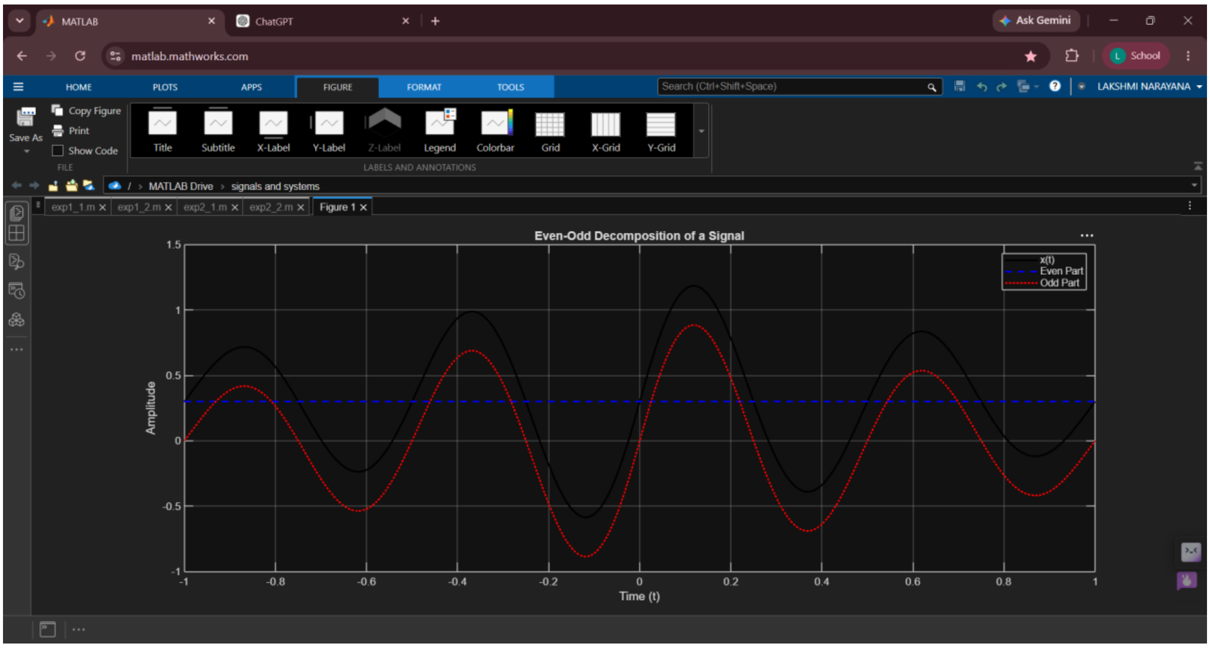
```
% Test Case: Addition and Multiplication of Two Sequences


clc;
clear;
close all;


% Define parameters
N = 30;                 % Number of samples
n = 0:N-1;              % Discrete time index vector
% Even-Odd Decomposition of a Signal


clc;
clear;
close all;


% Time vector
t = -1:0.001:1;
```

```
% Original signal
x = exp(-abs(t)).*sin(4*pi*t) + 0.3;

% Reverse signal x(-t)
x_rev = interp1(t, x, -t, 'linear');

% Even and Odd parts
xe = 0.5 * (x + x_rev);
xo = 0.5 * (x - x_rev);

% Plot the signals
figure;

plot(t, x, 'k', 'LineWidth', 1.5);
hold on;
plot(t, xe, 'b--', 'LineWidth', 1.5);
plot(t, xo, 'r:', 'LineWidth', 1.5);
grid on;

legend('x(t)', 'Even Part', 'Odd Part');
xlabel('Time (t)');
ylabel('Amplitude');
title('Even-Odd Decomposition of a Signal');
hold off;
% Define Sequence 1: Sinusoidal sequence
f1 = 0.05;            % Frequency in cycles/sample
A1 = 2;               % Amplitude
```

```
x1_n = A1 * sin(2 * pi * f1 * n);

% Define Sequence 2: Exponentially decaying sequence
a2 = 0.1;            % Decay constant
A2 = 3;              % Amplitude
x2_n = A2 * exp(-a2 * n);

% Addition of the two sequences
x_add = x1_n + x2_n;

% Multiplication of the two sequences
x_mult = x1_n .* x2_n;

% Plot the sequences and their results
figure;

% Plot Sequence 1
subplot(4,1,1);
stem(n, x1_n, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Sequence 1: Sinusoidal Sequence');

% Plot Sequence 2
subplot(4,1,2);
stem(n, x2_n, 'filled');
```

```
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Sequence 2: Exponentially Decaying Sequence');

% Plot the result of addition
subplot(4,1,3);
stem(n, x_add, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Addition of Sequence 1 and Sequence 2');

% Plot the result of multiplication
subplot(4,1,4);
stem(n, x_mult, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Multiplication of Sequence 1 and Sequence 2');
```

Output Waveform:



Result:

Signal equals sum of even and odd parts.